CAPSTONE PROJECT REPORT: CAR PRICE PREDICTION SYSTEM

Course Alignment: Honeywell AI/ML Certification Core Project

Academic Year: Second Year Engineering Capstone

1. Executive Summary

This project delivers an end-to-end Machine Learning solution designed to automate vehicle valuation. By evaluating structural vehicle metrics—including model year, age, distance driven, fuel mechanics, transmission type, and seller profile—the system fits a high-accuracy regression landscape to predict fair market value.

2. Methodology & Core Algorithm

Unlike traditional simple linear baselines, this architecture leverages an advanced **Random Forest Regressor** ensemble model. By building 100 distinct structural decision trees (`n_estimators=100`) over randomized bootstrap sub-samples of the historical data, the framework mitigates overfitting, balances out high-variance anomalies, and achieves robust generalisation boundaries on completely unseen test data sets.

3. Feature Engineering & Preprocessing Pipeline

• **Temporal Metric Synthesis:** Automatically converts static historical release years into an active continuous variable representing *Car Age* anchored to the year 2026.
• **Structural Reduction:** Drops unique identifier dimensions (`Car_Name`) to protect the structural array from sparse cardinal inflation.
• **Categorical One-Hot Encoding:** Transforms nominal text tracking attributes (e.g., Petrol vs. Diesel, Manual vs. Automatic) into discrete numeric array headers using dummy indexing parameters.

4. Core Implementation Script

```
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import r2_score, mean_absolute_error

# 1. Load the dataset
df = pd.read_csv('car_data.csv')

# 2. Data Preprocessing
df['Current_Year'] = 2026
df['Car_Age'] = df['Current_Year'] - df['Year']
df.drop(['Year', 'Current_Year', 'Car_Name'], axis=1, inplace=True)

# Convert text tracking fields into numbers
df = pd.get_dummies(df, drop_first=True)

# 3. Split features and target price
X = df.drop(['Selling_Price'], axis=1)
y = df['Selling_Price']
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

# 4. Train the Machine Learning Model
model = RandomForestRegressor(n_estimators=100, random_state=42)
model.fit(X_train, y_train)

# 5. Evaluate the model performance
predictions = model.predict(X_test)
print(f"Model R2 Score (Accuracy): {r2_score(y_test, predictions):.2f}")
print(f"Mean Absolute Error: {mean_absolute_error(y_test, predictions):.2f} Lakhs")
```

5. Pipeline Execution Framework Summary

Pipeline Stage	Operational Mechanics	Key Tool Used
Data Intake	Reads structural matrices from secondary raw storage.	`pandas.read_csv`
Preprocessing	Synthesizes vehicle age; performs categorical text dummy variable translation.	`pd.get_dummies`
Validation Split	Partitions dataset into 80% training matrix and 20% independent testing array.	`train_test_split`
Ensemble Fit	Builds and fits an array of 100 operational decision tree pathways.	`RandomForestRegressor`
Performance Audit	Validates predictive precision via R-squared alignment and Mean Absolute Error tracking.	`r2_score`, `mean_absolute_error`